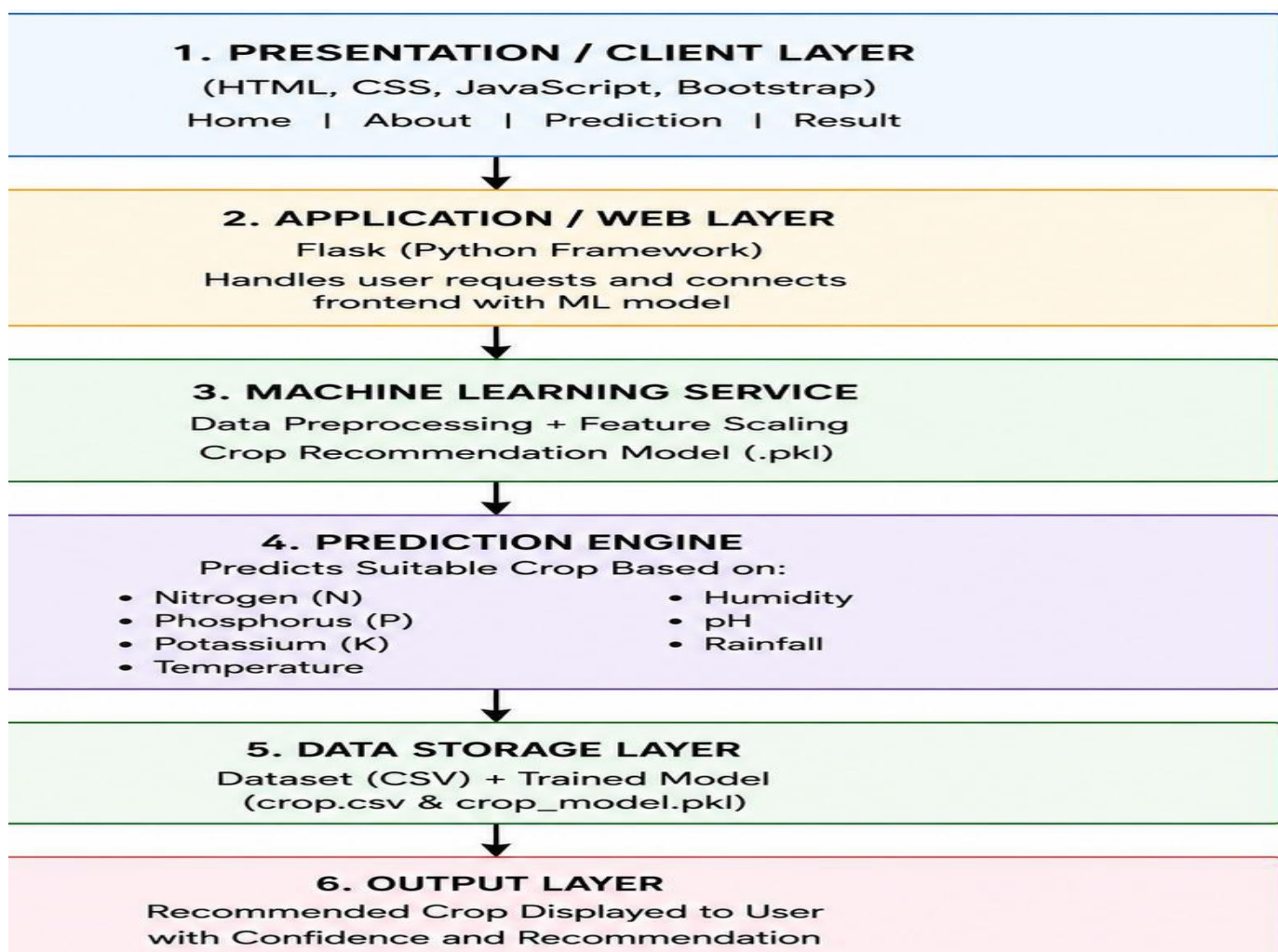


Solution Architecture

Date	12 July 2026
Team Id	XXXXXX
Project Name	Opti Crop: Smart Agricultural Production Optimization Engine
Maximum marks	5 marks

Solution Architecture diagram:



Component Name	Description / Role in Architecture	Technologies Used
Presentation Layer	Provides a user-friendly interface for farmers to input soil and environmental parameters and view crop prediction results.	HTML, CSS, Bootstrap, JavaScript
Application / Web Layer	Receives user input, validates the data, communicates with the machine learning model, and returns the prediction to the user.	Python, Flask
Machine Learning Service	Preprocesses the input values, loads the trained crop recommendation model, and predicts the most suitable crop.	Scikit-learn, NumPy, Joblib
Prediction Engine	Uses soil nutrients (N, P, K), temperature, humidity, pH, and rainfall values to generate an accurate crop recommendation.	Random Forest Classifier (ML Model)
Data Storage Layer	Stores the crop dataset used for training and the serialized machine learning model used for prediction.	CSV Dataset, Pickle (.pkl)
Output Layer	Displays the recommended crop along with the prediction result in an easy-to-understand format for the user.	Flask Templates (Jinja2), HTML